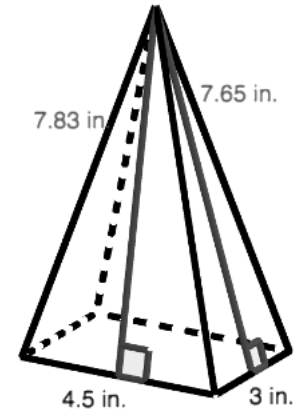
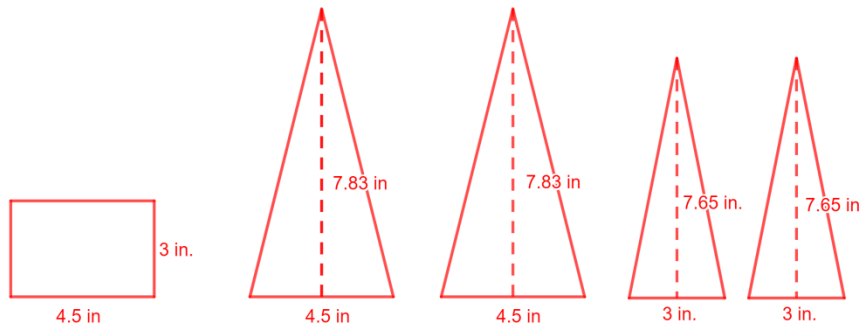


A rectangular pyramid is shown with measurements in inches (in.).

1. Draw a net or the individual faces of the figure. Label the measurements.

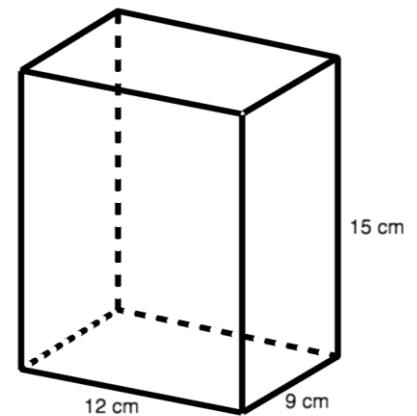
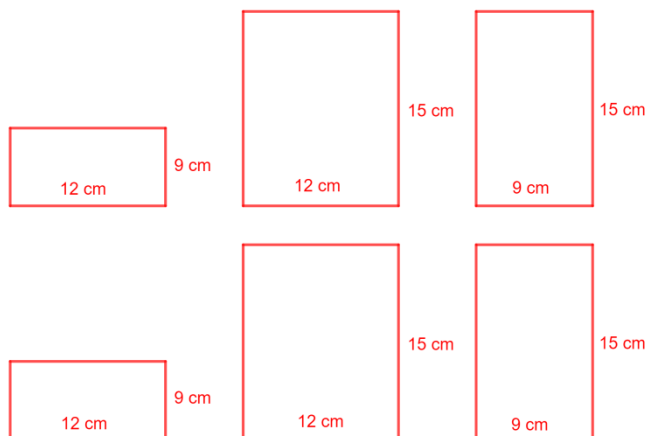


2. Find the surface area of the figure.

$$\begin{aligned} &135 + 2(0.5 \times 4.5 \times 7.83) + 2(0.5 \times 3 \times 7.65) \\ &= 13.5 + 2(17.6175) + 2(11.475) \\ &= 13.5 + 35.235 + 22.95 \\ &= 71.685 \text{ in.}^2 \end{aligned}$$

A rectangular prism is shown with measurements in centimeters (cm).

3. Draw a net or the individual faces of the figure. Label the measurements.



4. Find the surface area of the figure.

$$\begin{aligned} &2(108) + 2(180) + 2(135) \\ &= 216 + 360 + 270 \\ &= 846 \text{ cm}^2 \end{aligned}$$

5. Find the volume of the figure.

$$12 \times 9 \times 15 = 1,620 \text{ cm}^3$$

Sanjay is designing a suitcase. He drew a model of his suitcase.

6. If the case has a volume of 7,098 cubic inches (in^3), what is the length of the case?

$$\frac{7098}{(13 \times 28)} = \frac{7098}{364} = 19.5 \text{ in.}$$

7. How much material would Sanjay need to make a case with these dimensions, not taking into consideration any overlap of the faces?

$$2(19.5 \times 28) = 2(546) = 1,092$$

$$2(19.5 \times 13) = 2(253.5) = 507$$

$$2(13 \times 28) = 2(364) = 728$$

$$1,092 + 507 + 728 = 2,327 \text{ in}^2$$

8. If the inside of the case has a length of 18 in., a width of 11.5 in., and a height of 24 in., how much smaller is the volume of the inside of the case than the volume of the entire case?

$$7098 - (18 \times 11.5 \times 24) = 7098 - 4968 = 2,130 \text{ in}^3 \text{ smaller}$$



Jocelyn makes and sells bookends that are in the shape of rectangular pyramids.

9. What is the surface area of one bookend?

$$4 \times 4 = 16 \text{ in}^2$$

$$0.5 \times 4 \times 5.4 = 10.8 \text{ in}^2$$

$$16 + 4(10.8) = 59.2 \text{ in}^2$$

10. If Jocelyn paints only the triangular faces, what is the surface area of the area she wants to paint?

$$4(10.8) = 43.2 \text{ in}^2$$

